



OPEN Spatiotemporal patterns and environmental determinants of wildlife-vehicle collisions in Banke National Park, Nepal

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Wildlife vehicle collisions (WVCs) represent a significant threat globally to biodiversity conservation, driver safety and economic resources. This study evaluated the magnitude, spatial-temporal patterns, and influencing factors associated with WVCs within Banke National Park (BaNP), Nepal, to inform effective mitigation planning and biodiversity conservation strategies. Data on WVCs and relevant environmental and anthropogenic variables were collected between April 2015 and March 2024 along a 97.3 km road segment traversing BaNP. The methodological approach integrated field surveys, spatial analyses using kernel density estimation, and statistical modelling to pinpoint collision hotspots and elucidate contributing factors. In total, 488 WVC incidents were documented, predominantly involving mammals (87.52%), followed by reptiles (8.58%) and birds (3.88%). Endangered and vulnerable species, including the golden monitor lizard, leopard cat, and four-horned antelope, were among the recorded fatalities. Spatial analysis identified significant WVC hotspots at *Muguwa*, *Obhari*, and *Khairi*, collectively accounting for over 60% of total incidents. Temporal patterns indicated a seasonal peak during autumn (30%), coinciding with increased post-monsoon dispersal and foraging behaviour. Statistical analyses highlighted that road curvature, proximity to water sources, and surrounding land use significantly influenced collision risk. Specifically, higher WVC odds were associated with curved roads and greater distances from settlements, while straighter road segments and higher forest canopy coverage near roads correlated with reduced risk. This study underscores the pressing need for targeted mitigation measures, such as wildlife crossings, speed reduction zones, and enhanced public awareness initiatives, to reduce WVC occurrences and protect vulnerable wildlife in BaNP. The findings contribute to the existing knowledge regarding road ecology and provide crucial insights for informed conservation interventions within protected areas.

Keywords Road kill hotspots, Road ecology, Conservation, Mitigation, Environmental factors

Wildlife-vehicle collisions (WVCs) represent a pervasive global challenge, threatening biodiversity, human safety, and economic resources across six continents. From North America's highways fragmenting ungulate migrations to Southeast Asia's roads encroaching on tiger habitats, the ecological impacts of transportation networks are universal^{1–3}. WVCs occur when animals collide with vehicles on roads, posing significant risks to both wildlife and motorists. This phenomenon is driven by the simultaneous presence of wildlife and vehicular traffic, characterized by the unpredictable behaviour of both parties^{4,5}. WVCs represent a global concern, leading to financial losses, species conservation threats, and driver safety hazards⁶. The presence of wildlife on roads

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is influenced by a complex interplay of factors including road characteristics, traffic flow, landscape features, weather conditions, and animal behaviour⁷. Wildlife often congregates near roadside ditches where grass and water are plentiful, particularly when alternative food and water sources are scarce. Dumping garbage along the roadside, dropping food by vehicles, and provisioning from passing vehicles can also attract wildlife as in the case of nonhuman primates^{8,9}. Roads also provide open spaces for social interactions and facilitate access to new territories for dispersing young individuals^{6,10–12}.

Animals frequently cross roads in search of crops, pastures, water, or territories⁵. In protected areas, roads significantly impact the natural environment and ecosystem health. One of the most obvious indicators of this impact is wildlife mortality on roads¹³. Highways that cross wildlife refuges are particularly disruptive, adversely affecting wildlife and their habitats¹⁴. The consequences include altered movement patterns, changes in feeding and movement behaviours¹⁵, and mortality from vehicle collisions^{4,16,17}. Wildlife species such as deer exhibit a “freeze” response commonly known as ‘deer in the headlights’ to avoid predator detection, unfortunately, this response does not work for moving vehicles^{18,19}. Collectively, highways can have an unsettling effect on the ecological balance and genetic diversity of wildlife populations²⁰.

Wildlife mortality due to vehicle collisions is not randomly distributed along roads but is spatially clustered because of the relationships between wildlife movements and specific habitats, terrains, and adjacent land use types^{21–23}. Therefore, landscape spatial patterns are crucial in determining areas where the likelihood of WVC is relatively high^{23,24}. Regardless of their size, roads function as barriers that impede the natural movement of animals²⁵. Large animals with large home ranges or species that disperse widely are particularly susceptible to encountering these barriers²⁶. This impact is especially pronounced for rare and threatened species or those with low population densities²⁷.

Mitigation measures for wildlife-vehicle accidents are vital for reducing biodiversity threats and enhancing human safety^{28,29}. Effective strategies include wildlife underpasses, jump-out structures, and exclusion fencing, which have proven successful in decreasing collisions between vehicles and wildlife³⁰. Additionally, roadside animal detection systems have been implemented to reduce WVC and improve habitat connectivity⁷. Analytical frameworks have also been developed to identify the factors influencing wildlife-vehicle collisions and their spatial-temporal patterns, aiding in formulating effective mitigation strategies⁶. It is important to minimise the negative impacts of highways on protected areas and wildlife²⁰. The implementation of ecologically-based prioritization frameworks can help to identify road sections with a greater likelihood of WVC, thus enabling targeted deployment of mitigation measures³¹. Additionally, identifying wildlife corridor intersections can guide the placement of permanent mitigation measures^{32,33}, thereby reinforcing locations prone to roadkill incidents³⁴.

Over the past few decades, the detrimental effects of roads on wildlife populations have gained global recognition^{35,36}. In Nepal, WVCs have become a significant concern, and research conducted along the country's East-West (Mahendra) Highway and Chitwan National Park has shown that species such as spotted deer (*Axis axis*) and oriental garden lizards (*Calotes versicolor*) frequently fall victim to these collisions³⁷. Retrospective regional and global studies have revealed that some carnivore taxa, such as Artiodactyla, Felidae and Herpestidae are particularly vulnerable to mortality from vehicle collisions, significantly threatening native mammal populations^{2,38}.

The East-West Highway, which runs through various protected areas in Nepal, has experienced frequent WVCs thus jeopardizing conservation efforts in a country that has made notable progress in recent years. Among these protected areas, Banke National Park (BaNP) has recorded the highest number of such accidents where road expansion intersects critical biodiversity hotspots. WVCs exemplify a crisis mirrored in developing nations racing to balance infrastructure growth with conservation³⁹.

Therefore, this study aims to assess the magnitude, patterns, and factors associated with WVCs in BaNP, providing critical information for mitigation planning and conservation efforts. This study further provides a microcosm of both local and global road ecology challenges, and offers scalable insights into hotspot prediction, species vulnerability, and mitigation strategies applicable to protected areas.

Methods and methodology

Study area

This study was conducted along the highways of Banke National Park (BaNP), which is located in the western region of Nepal (Fig. 1). The BaNP spans geographical coordinates 81°39'29" and 82°12'19" east longitude and 27°58'13" to 28°21'26" north latitude and elevational ranges from 100 m above sea level (masl) to 1200 masl. The Park covers an area of 550 sq. km² (with an additional buffer zone of 343 sq. km² and serves as a critical habitat for the endangered Bengal tiger (*Panthera tigris tigris*), a flagship species of the area⁴⁰. BaNP also forms part of a transboundary landscape, connecting with the Suhelwa Wildlife Sanctuary in India through national and community forests to the south. Additionally, the Park shares borders with Nepal's Bardia National Park to the west, which, in turn, links to the Katarniaghat Wildlife Sanctuary in India through the Khata corridor, national forests, and community forests.

BaNP is characterized by diverse biodiversity, including 124 plant species, 34 species of mammals, over 300 bird species, 24 reptile, 7 amphibian, and 58 fish species. The natural forest cover is predominantly composed of Sal (*Shorea robusta*), Karma (*Haldina cordifolia*), Khair (*Acacia catechu*), and Sissoo (*Dalbergia sissoo*)⁴¹.

The study area of focus consisted of a 97.3 km stretch of road within BaNP, which included 71.6 km of the East-West Highway from Shiva-Khola in the east to Kohalpur, and a 25.7 km section of the Ratna Highway from Kohalpur to Deurali (Fig. 1). These roads traverse critical wildlife habitats and are known for frequent WVCs.

Data collection

Data collection covered the period from April 2015 to March 2024. The authors (NPK, AK and KD) collected data on the natural and anthropogenic variables from the field and WVC data from Banke National Park and

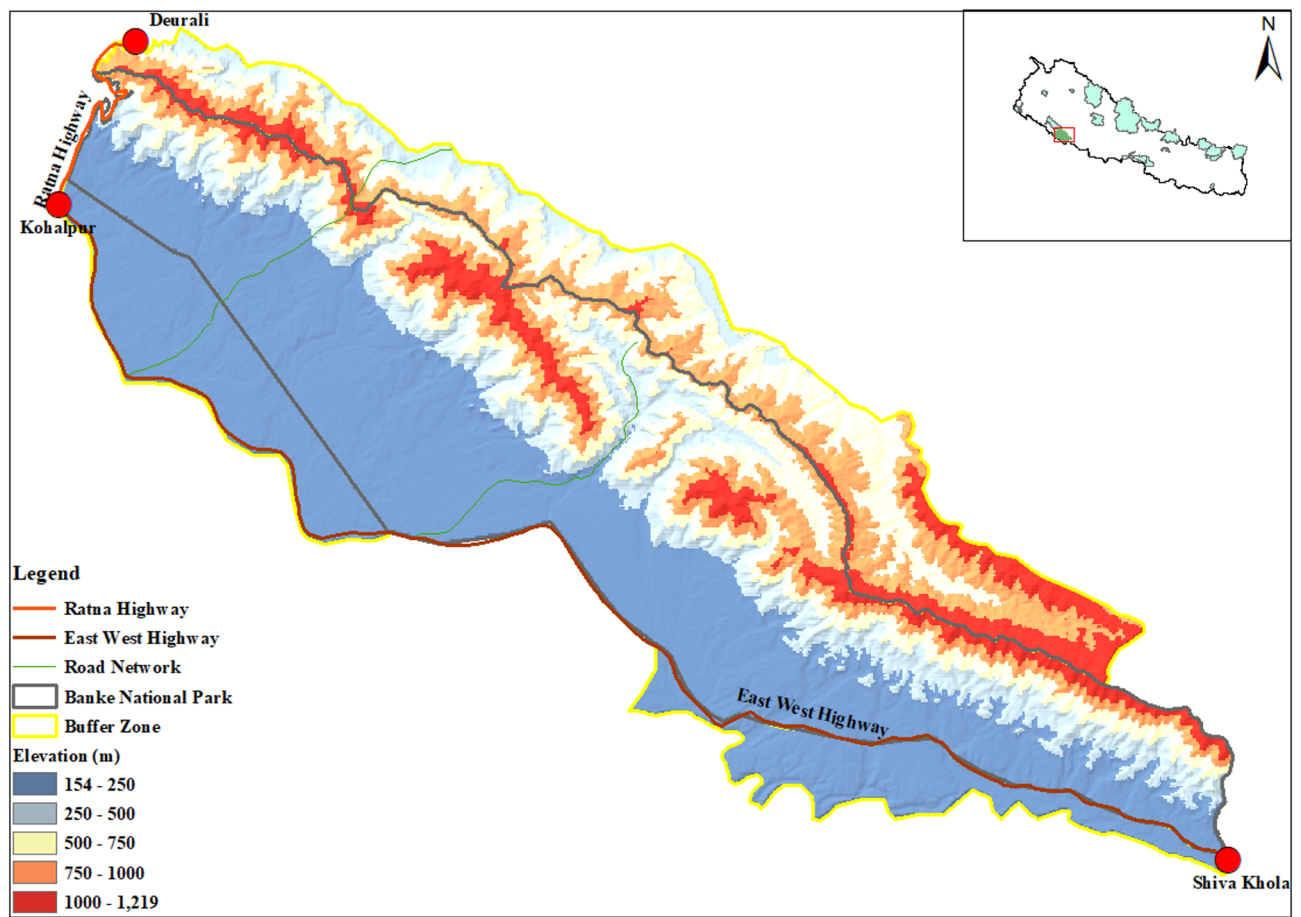


Fig. 1. Map of Banke National Park and surrounding area. The black line depicts the East-West (Mahendra) and Ratna Highways passing through and along the western border of the park, respectively. Data source: Department of Survey, DNPWC/MoFSc, Government of Nepal.

the Department of National Park and Wildlife Conservation. The 97.3 km stretch of road for the targeted study was divided into 500-meter segments to ensure consistent spatial resolution for field data collection and analysis. This resolution was chosen on the basis of consistency with previous studies where 500-meter segments are commonly used in road ecology to balance spatial resolution with statistical robustness, ecological relevance and logistic feasibility^{6,42}. This approach allows for standardized assessment of environmental and anthropogenic variables and enables detailed spatial modelling of WVC patterns. The environmental variables included natural landscape features such as vegetation attributes (i.e., crown cover, forest type), proximity to water sources, and land use classified as forested areas. Similarly, anthropogenic variables encompass human-modified factors such as road characteristics (curvature, straightness, surface type), proximity to settlements, and infrastructure elements (bridges, culverts, and road signs). Road traffic data for the study area are not systematically maintained or publicly available, thus limiting our ability to include this parameter in the analysis.

In the study area, most of the road segments fell within the core area of BaNP. Some segments also intersected with the buffer zone, especially along the East-West Highway stretch between Kohalpur and Deurali. In addition, infrastructure elements were documented by noting the presence of wildlife signs, bridges, and culverts. Along the 71.6 km stretch of the East-West Highway, a total of 39 bridges, 59 culverts, and 59 road signs were identified. However, because very few WVC incidents occurred on the Ratna Highway, the impact of the infrastructural features was evaluated only for the East-West Highway between Shiva Khola and Kohalpur. Finally, the entire 97.3 km stretch of road was further subdivided into 19 locations based on local names, with each location spanning 4–6 km.

Data on WVCs were compiled from records maintained by the BaNP authorities [DNPWC staff, security personnel, including authors] from April 2015 to March 2024. Additionally, from February 2021 to June 2022 and April–May 2024, the authors collected additional field data, including observations of WVC incidents and associated environmental parameters. The data on WVCs were categorized by season—spring/premonsoon (March–May), summer/monsoon (June–September), autumn/postmonsoon (October–November), and Winter (December–February)⁴³—and by time of day: morning (04:00–10:00), daytime (10:00–16:00), evening (16:00–22:00), and night (22:00–04:00)^{44,45}.

Data analysis

To analyze the influence of environmental and anthropogenic factors on WVCs, we employed two complementary generalized linear modelling approaches using R software (version 4.2.1)⁴⁶. First, we fitted a negative binomial GLM, implemented with the `glm.nb` function in the MASS package⁴⁷, to the aggregated data across multiple locations. This “aggregated model” was designed to capture overarching patterns and the combined effects of predictor variables while explicitly accounting for interlocation variability and overdispersion in the count data. For comparison, we also fitted separate GLMs, assuming Gaussian error distributions, to data subsets from individual locations, thereby providing more precise estimates of localized impacts on WVC risk factors^{48,49}. In both cases, appropriate link functions (log-link for the negative binomial model and identity-link for the Gaussian models) were used, and model selection and averaging were guided by Akaike’s information criterion (AIC), with predictors retained in the final models only if they significantly improved fit ($\Delta AIC < 2$). Where the data structure warranted, mixed-effects models were additionally explored using the `lme4` package⁵⁰. This dual modelling strategy ensured that both broad-scale trends and fine-scale, location-specific influences on WVCs were comprehensively evaluated.

The model predictors included road characteristics (e.g., curvature, surface type, and straightness), vegetation attributes (e.g., crown cover and forest type), land use (forested versus settlement areas), and proximity to key features (e.g., distances from settlements and water sources). Prior to analysis, continuous predictors were log-transformed when necessary to meet the assumptions of linearity and homoscedasticity and then standardized (z-transformed) to facilitate direct comparisons of effect sizes. Multicollinearity among predictors was assessed using variance inflation factors (VIFs), with any variable exhibiting a $VIF > 5$ excluded from the final models. Model diagnostics included residual analyses (e.g., Q–Q plots, fitted versus residual plots) to verify the assumption, and 95% confidence intervals were calculated for all parameter estimates to assess robustness. Visualizations of the model outputs were generated using the `ggplot2` package⁵¹. All the statistical tests were two-tailed and considered significant at $p < 0.05$, and model diagnostics confirmed that the GLMs adequately met the underlying assumptions.

Temporal analysis

The chi-square test for goodness of fit was used to assess the relationship between the WVC frequency with seasonal or temporal patterns. The test evaluated whether the distribution of WVCs across seasons and time was statistically significant.

Hotspot analysis

To identify WVC hotspots, we used a kernel density estimation^{52,53}. Kernel density estimates areas of high and low intensity by measuring the dispersion pattern of locations of interest within a given area. The density plots were generated using ‘`stat_density2d`’ from `ggplot2`, which internally calls the ‘`kde2d`’ function of the MASS package to estimate and smooth the two-dimensional probability density of data points. In this case, we output the results in filled contour bands, where areas of different density levels are depicted with varying colours—high (yellow) to low (dark purple). The hotspot assessment was conducted for four species that had the highest record of WVCs in the study area: wild boar, spotted deer, jackals, and rhesus monkeys.

Results

Species susceptible to WVC

Over the 10-year study period (April 2015 - March 2024), a total of 488 animals were recorded as killed in WVCs in Banke National Park. Mammals accounted for the majority of WVC incidents (87.5% of the total), followed by reptiles (8.6%) and birds (3.9%). In total, 20 mammal species, seven bird species, and four reptile species were identified as victims of WVC. The most frequently killed species included the wild boar (*Sus scrofa*), with 92 incidents (18.8% of total WVC), rhesus monkey (*Macaca mulatta*), with 72 incidents (14.7%), spotted deer (*Axis axis*), with 62 incidents (12.7%), and jackal (*Canis aureus*), with 49 incidents (10.0%). Other species with notable WVC frequencies included the Indian crested porcupine (*Hystrix indica*), barking deer (*Muntiacus vaginalis*), jungle cat (*Felis chaus*), and flying squirrel (*Hylopetes* sp.), each with 16–23 incidents (Table 1).

Several species of conservation concern were also affected by WVCs. These included the endangered golden monitor lizard (*Varanus flavescens*), the vulnerable four-horned antelope (*Tetracerus quadricornis*), leopard cat (*Prionailurus bengalensis*), and sambar deer (*Rusa unicolor*), as well as Near Threatened species such as the Bengal monitor lizard (*Varanus bengalensis*), Indian rock python (*Python molurus*), and gray langur (*Semnopithecus hector*). Five species were listed under CITES Appendix I, two under Appendix II, and seven under Appendix III, emphasizing the conservation significance of the affected wildlife (Table 1).

Spatial distribution of wildlife vehicle collisions

The spatial distribution of WVCs along the highway was not uniform, with significant clustering observed in specific locations (Fig. 2). The areas of Muguwa, Obhari, Khairi and Samsergunj were identified as WVC hotspots, accounting for over 60% of all recorded WVCs. Muguwa (103 incidents, 21.1% of total WVCs), followed by the Obhari (98 incidents, 20.0%), Khairi (66 incidents, 13.5%) and Samsergunj (31 incidents, 6.3%).

The temporal analysis of WVCs revealed that the year 2022 had the highest number of incidents (18.2% of total WVCs), followed by 2017 (17.4%), 2018 and 2016 (10.8%). The lowest number of WVCs was recorded in 2020 (7.4%), likely due to reduced vehicular traffic during the COVID-19 pandemic (Fig. 3).

Species-specific WVC hotspot analysis

Species-wise WVC hotspot mapping revealed distinct spatial patterns for the four (see Table 2) most frequently killed species: wild boars, rhesus monkeys, spotted deer and jackals. Of the total wildlife killed during the study

Index	Common name	Scientific name	IUCN status	CITES status	NPWC Act	Total number(2015–2024)
1.	Wild boar	<i>Sus scrofa</i>	LC			92
2.	Rhesus monkey	<i>Macaca mulatta</i>	LC	II		72
3.	Spotted deer	<i>Axis axis</i>	LC			62
4.	Jackal	<i>Canis aureus</i>	LC	III		49
5.	Indian crested porcupine	<i>Hystrix indica</i>	LC			23
6.	Barking deer	<i>Muntiacus vaginalis</i>	LC			20
7.	Jungle cat	<i>Felis chaus</i>	LC	III		16
8.	Flying squirrel	<i>Hylopetes sp</i>	LC			16
9.	Large Indian civet	<i>Viverra zibetha</i>	LC			15
10.	Four horned antelope	<i>Tetracerus quadricornis</i>	VU	III	P	15
11.	Golden Monitor lizard	<i>Varanus flavescens</i>	EN	I	P	13
12.	Dove	Not identified				11
13.	Snake	Not identified				10
14.	Bengal monitor lizard	<i>Varanus bengalensis</i>	NT	I		9
15.	Leopard cat	<i>Prionailurus bengalensis</i>	LC	II	P	9
16.	Leopard	<i>Panthera pardus</i>	VU	I		8
17.	Python	<i>Python molurus</i>	NT	I	P	7
18.	Squirrel	Not identified				7
19.	Mongoose	Not identified				5
20.	Gray langur	<i>Semnopithecus hector</i>	NT	I		4
21.	Hare	<i>Lepus nigricollis</i>	LC			8
22.	Yellow throated marten	<i>Martes flavigula</i>	LC	III		3
23.	Blue bull	<i>Boselaphus tragocamelus</i>	LC	III		2
24.	Sambar deer	<i>Rusa unicolor</i>	VU			2
25.	Eagle	Not identified				2
26.	Unidentified bird	Not identified				2
27.	Drongo	Not identified				2
28.	Hyena	<i>Hyaena hyaena</i>	NT	III	P	1
29.	Peacock	<i>Pavo cristatus</i>	LC	III		1
30.	Myna	Not specified				1
31.	Common moorhen	<i>Gallinula chloropus</i>				1
	Total frequency					488

Table 1. Frequency and conservation status of wildlife species killed in wildlife-vehicle collisions (WVCs) along the E-W highway in Banke National park (BaNP), Nepal. LC = least concern, nt = near threat, v = vulnerable, e = endangered Conservation status is shown across three criteria: international union for conservation of nature (IUCN), the convention on international trade in endangered species of wild fauna and flora (CITES), and Nepal's National parks and wildlife conservation act, 2029 (1973).

period, these four species accounted for 56% of the individuals killed (Fig. 4A). For wild boars, the high-fatality area extended from Samserganj to Obhari, with 80 incidents, whereas the medium-fatality areas were identified in Maguwa (15 incidents) and Khairi (14 incidents). Collectively, these areas accounted for 67.4% of all wild boar collisions. For rhesus monkeys, the high-fatality area was located between Obhari and Maguwa, contributing to 50% of the total collisions for this species. Other hotspots included Khairi and Samsergunj. Spotted deer collisions were concentrated in four major hotspots, with the high-risk area between Dhakeri and Obhari, where the risk increased from west to east. The Maguwa to Khairi areas presented medium to high risk, whereas Pragatinagar and Shivakhola presented medium risk. For jackals, the high-fatality area was in Samsergunj, with medium-risk areas in Khairi and the low-risk regions in Pragatinagar [Figure 4B; Supplementary Table 1].

Temporal patterns in WVCs

A temporal analysis revealed significant seasonal and diurnal variations in WVCs. Seasonally, the highest number of collisions occurred during autumn, accounting for 30% of total WVC, followed by spring (27%), winter (26%), and summer (17%). The chi-square test confirmed that WVC significantly depended on the season ($p = 0.001$).

Species-specific seasonal patterns were also observed in wild boars, rhesus monkeys, and spotted deer ($p\text{-value} < 0.05$). Wild boar collisions were highest in spring (39.1%), rhesus monkey collisions peaked in autumn (41.1%), and spotted deer collisions were highest in spring (41.9%) (Table 2).

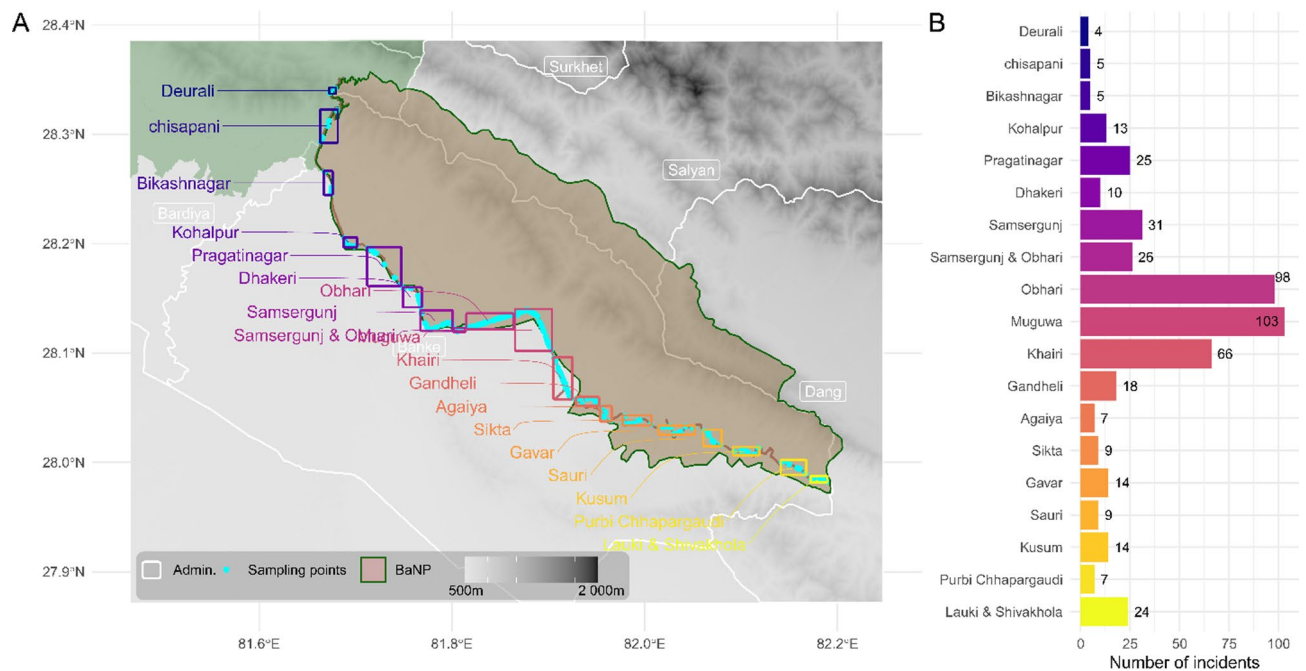


Fig. 2. The spatial distribution of wildlife-vehicle collisions (WVCs) at different locations along the E-W highway in Banke National Park (BaNP) (A), and location-wise summary of incidents (B). The rectangular bounding extents around sampling points (cyan-coloured dots) in A demark different locations and are coloured to match the bar plot to improve visualization.

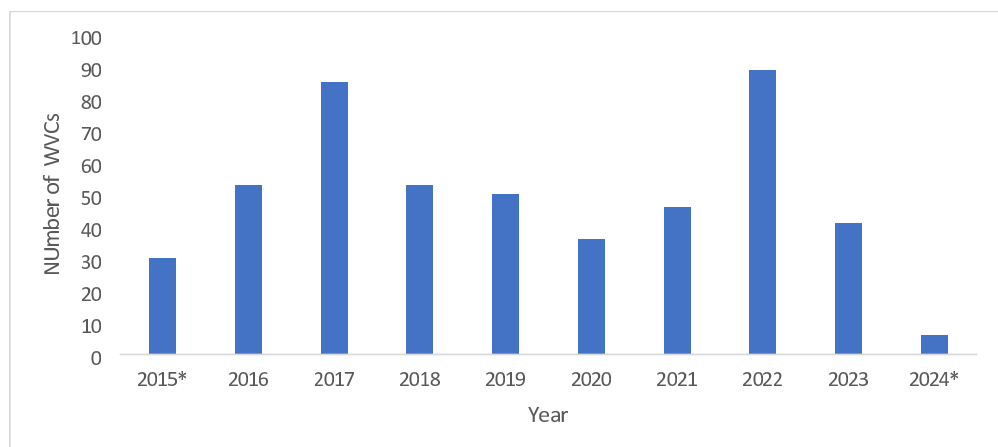


Fig. 3. The temporal variation in the number of wildlife-vehicle collisions (WVCs) along the E-W highway in Banke National Park (BaNP) during the study period. *Partial data for the year.

Species common name	Spring	Summer	Autumn	Winter	p-value
Wild boar	39.1	16.3	16.3	28.3	0.004
Rhesus monkey	8.2	12.3	41.1	38.4	0.001
Spotted deer	41.9	19.4	11.3	27.4	0.005
Jackal	18.4	18.4	32.7	30.6	0.061

Table 2. Seasonal variation in the number (percentage) of wildlife-vehicle collisions along the E-W highway in Banke National park (BaNP) for the top five most affected species. The highest frequencies and statistically significant differences between seasons are shown in bold fonts.

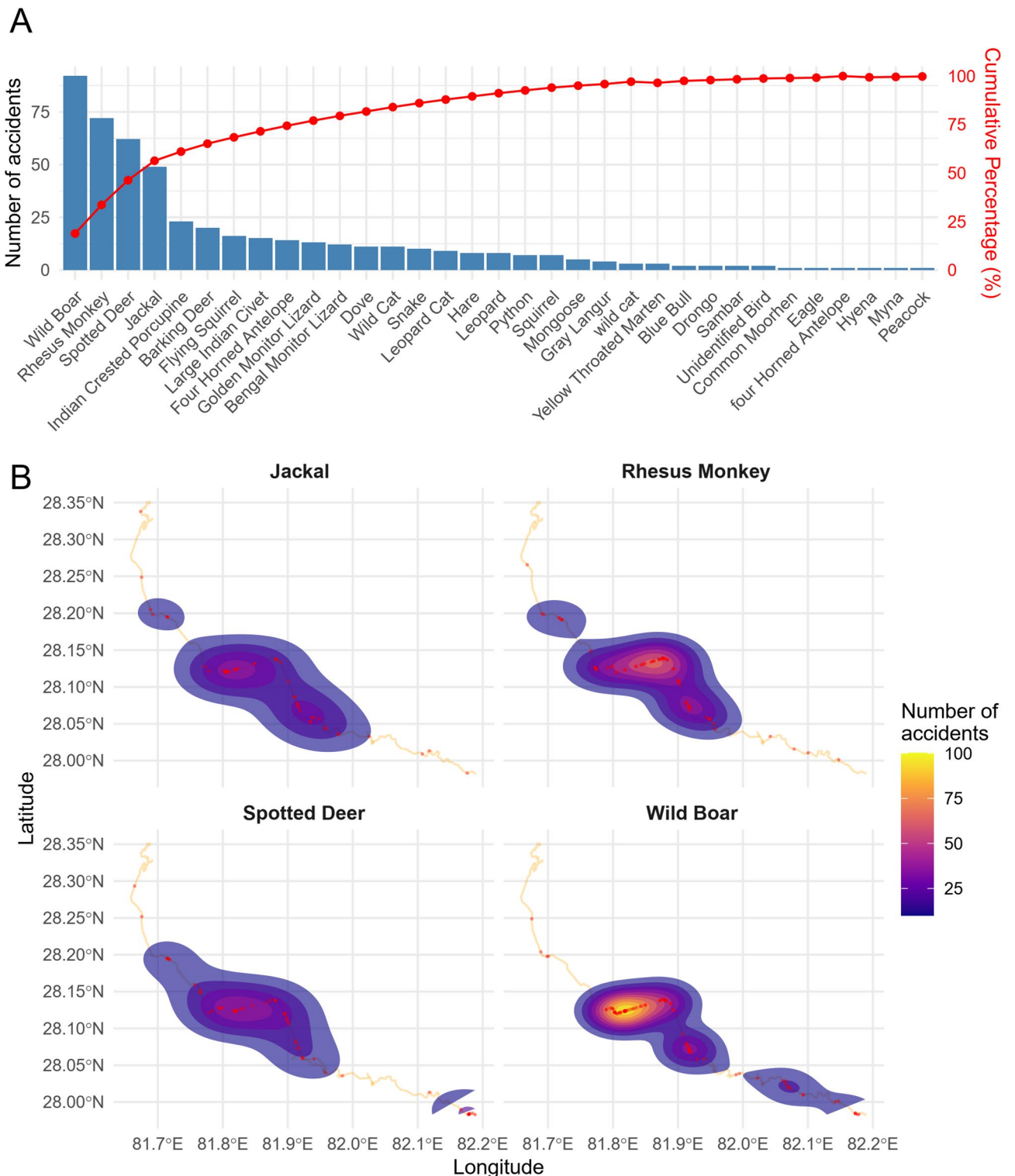


Fig. 4. Species-specific cumulative percentage of wildlife-vehicle collisions (WVCs) (A), distribution of WVC hotspots for wild boars, rhesus monkeys, spotted deer, and jackals along the E-W highway in Banke National Park (BaNP) (B).

The highest number of WVC events occurred during the morning shift (32%), followed by the night shift (28%), the evening shift (24%), and the day shift (16%). The chi-square test confirmed a statistically significant relationship between time of day and WVC ($\chi^2 = 23.095$, $df = 3$, $p = 0.00039$). Species-specific temporal patterns were also substantial for wild boars, rhesus monkeys, Indian crested porcupines, and golden monitor lizards ($p < 0.05$) (Table 3).

Factors affecting wildlife-vehicle collisions

Key environmental and anthropogenic factors significantly influenced WVC risk across the study area (Fig. 5A, B). Significant spatial variation was observed. Locations such as *Maguwa*, *Obhari*, *Samsergunj*, *Kusum*, *Pragatinagar*, and *Shivakhola* presented the strongest positive associations with WVC risk, as indicated by positive coefficient estimates and elevated log means. However, areas such as *Kohalpur*, *Sikta*, and *Gabahar* presented a reduced risk, reflected in negative coefficient values. These patterns suggest that the influence of predictor variables on collision risk varies substantially among different regions.

Further analysis identified several specific predictors that modulate WVC risk. Road curvature emerged as a critical factor. Segments with high curvature (i.e., > 75% curvature) were significantly associated with increased collision risk, likely due to reduced driver visibility and delayed reaction times. Conversely, road sections characterized by high straightness (i.e., > 30% straightness) were linked to lower collision probabilities. Vegetation attributes also played a vital role. Areas with moderate to dense crown cover (50–75%) were positively associated with WVC, whereas sparse canopy cover (<25%) corresponded to reduced risk. Similarly, roads traversing forested land presented greater collision risk than did those in settlement areas, underscoring the importance of proximity to wildlife habitats. In addition, roads located near key landscape features presented altered risk levels. Specifically, proximity to settlements (100–500 m) and water sources (500–1000 m) was significantly associated with increased WVC risk, highlighting the influence of edge effects and resource attractors on wildlife movement. Regions dominated by Sal forests (with >60% coverage) consistently presented positive associations with WVC risk, likely reflecting the suitability of these habitats for wildlife that intersect road networks (Fig. 5A and B, Supplementary Tables 2 and Supplementary Table 3).

Impact of highway infrastructure on WVC

The study also evaluated the influence of highway infrastructure (i.e., road signs, bridges, culverts), on the occurrence of WVC. Road signs were associated with the highest median accident counts, with collisions occurring most frequently within 50–100 m of these markers. Notably, these locations exhibited high variability in accident counts, including several outliers where counts exceeded 20 incidents, suggesting that areas near road signs may serve as hotspots for WVC. This pattern may reflect the increased likelihood of wildlife crossing near these zones or reduced driver caution in such areas.

In contrast, culverts were linked to the lowest median accident counts, and the distances from culverts to WVC sites were relatively consistent, with a median distance of approximately 150 m. This observation suggests that culverts might function as effective safe crossing structures for wildlife, thereby mitigating collision risk. Bridges presented moderate accident counts with greater variability in the distance to collision sites (median ~ 200 m), indicating that WVC tend to occur farther from bridges than road signs (Fig. 6). These findings underscore the need to incorporate the spatial characteristics of highway infrastructure into WVC mitigation strategies.

Discussion

The findings from this study of wildlife-vehicle collisions in Banke National Park, Nepal, provide critical insights into the ecological and conservation challenges posed by roads inside protected areas. This study highlights both the spatial and temporal dimensions of WVC, identifying key wildlife species at risk, hotspot locations, and potential environmental factors contributing to road kills. These results align with previous studies that have demonstrated the severe ecological consequences of roads on wildlife, particularly through vehicle-induced mortality, habitat fragmentation, and behavioural disruptions^{36,54}.

Species susceptible to WVC

The study recorded 488 individual WVCs over ten years, with mammals accounting for the highest proportion (87.5%), followed by reptiles (8.6%) and birds (3.9%). These patterns are consistent with other studies, where mammals often constitute the majority of WVC incidents because of their larger size and mobility^{15,55–57}. Notably, several species of conservation concern were impacted, including the endangered golden monitor lizard, the vulnerable four-horned antelope, and near-threatened species such as the Bengal monitor lizard.

S.N.	Species	Total	χ ²	P value
1	Wild boar	92	12.489	0.005
2	Rhesus monkey	72	14.815	0.018
3	Spotted deer	62	21.167	0.001
4	Jackal	49	4.4	0.221
5	Indian crested porcupine	23	13.556	0.003
6	Barking deer	20	5.5	0.138
7	Jungle cat	16	24.733	0.001
8	Flying squirrel	16	3.333	0.343
9	Large Indian civet	15	0.667	0.880
10	Four horned antelope	15	5.364	0.147
11	Golden monitor lizard	13	10.455	0.015

Table 3. Species-wise temporal shifts (time of day) wildlife-vehicle collisions (WVC) pattern along the E-W highway in Banke National park (BaNP). The statistically significant differences are shown in bold fonts.

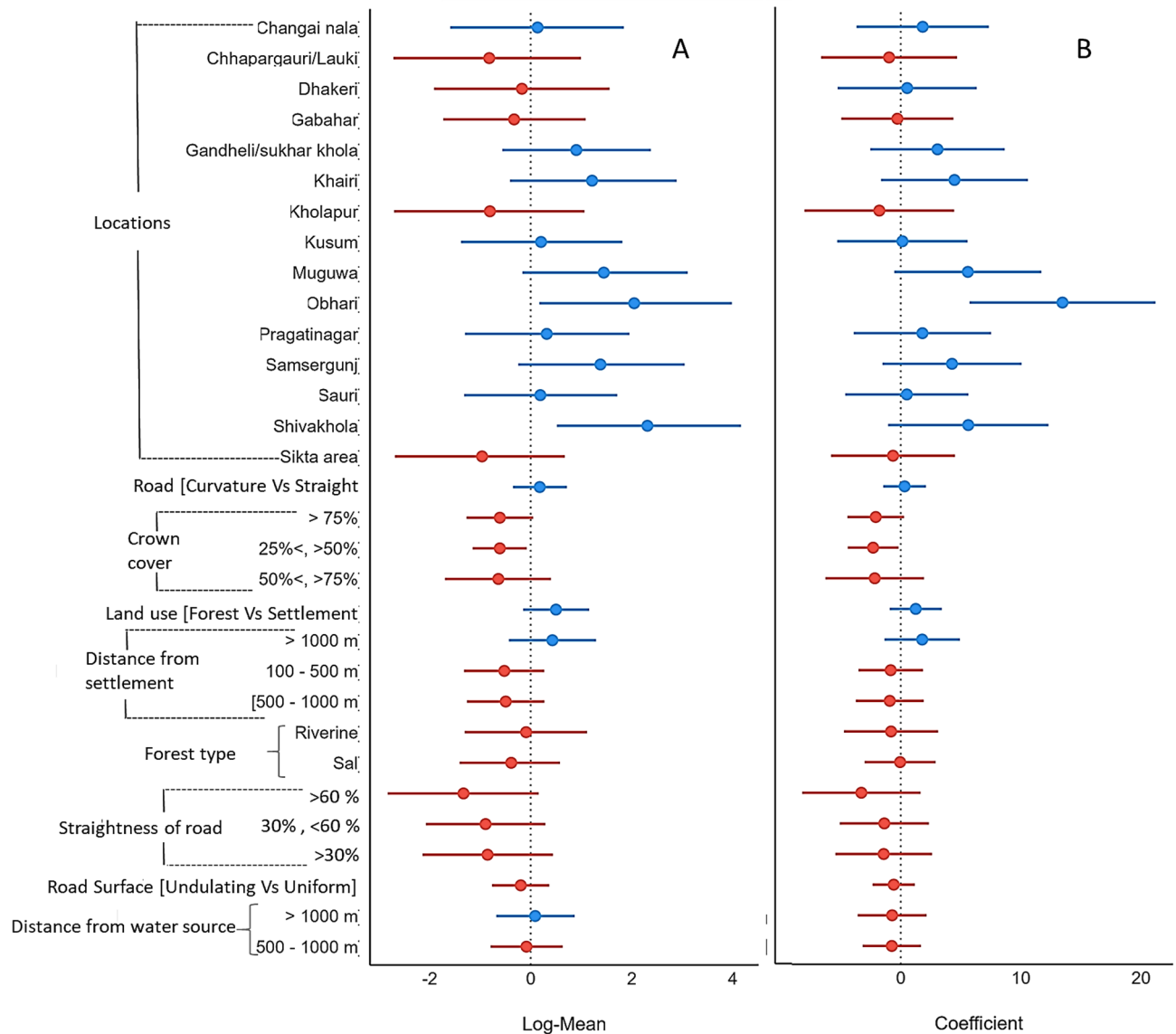


Fig. 5. Factors affecting wildlife-vehicle collisions (WVC) along the E-W highway in Banke National Park (BaNP). The effects were estimated using negative binomial (A) and standard generalized linear models (B). The red mean (point) + SD (wings) parameters denote negative effects, while the blue means denote positive effects.

These findings highlight the threat that roads pose to already vulnerable and endangered species, thus increasing their risk of local extinction^{58–60}.

Wild boars, rhesus monkeys, and spotted deer and jackals were recorded as the most frequently killed species. This pattern is consistent with other studies that have shown that large mammals are often above average in WVC records because of their greater visibility and movement across open landscapes^{55,61–64}. Hill, et al.⁶⁵ explained that wildlife uses roads for communication, foraging, movement, refuge, thermoregulation, and foraging. The high percentage of these species in WVC may also reflect their abundance and habitat preferences, with species such as the wild boar being particularly adaptable to fragmented landscapes and living in groups^{64–66}.

WVC hotspots and temporal patterns

The clustering of WVC incidents in specific areas, particularly in Khairi, Muguwa, and Obhari, indicates the presence of WVC hotspots, where collisions are disproportionately greater than in other areas. Hotspots are commonly associated with certain factors, such as road curvature, forest cover, proximity to water sources, and human settlement from highways, as also observed in studies from other regions^{67,68}. Identifying these hotspots provides a basis for targeted mitigation strategies, such as installing wildlife crossing structures or modifying road designs to reduce collision risk^{69,70}.

The temporal patterns of WVC in BaNP showed seasonal variation, with the highest number of collisions occurring during the autumn months (30%), followed by spring (27%) and winter (26%). This seasonality

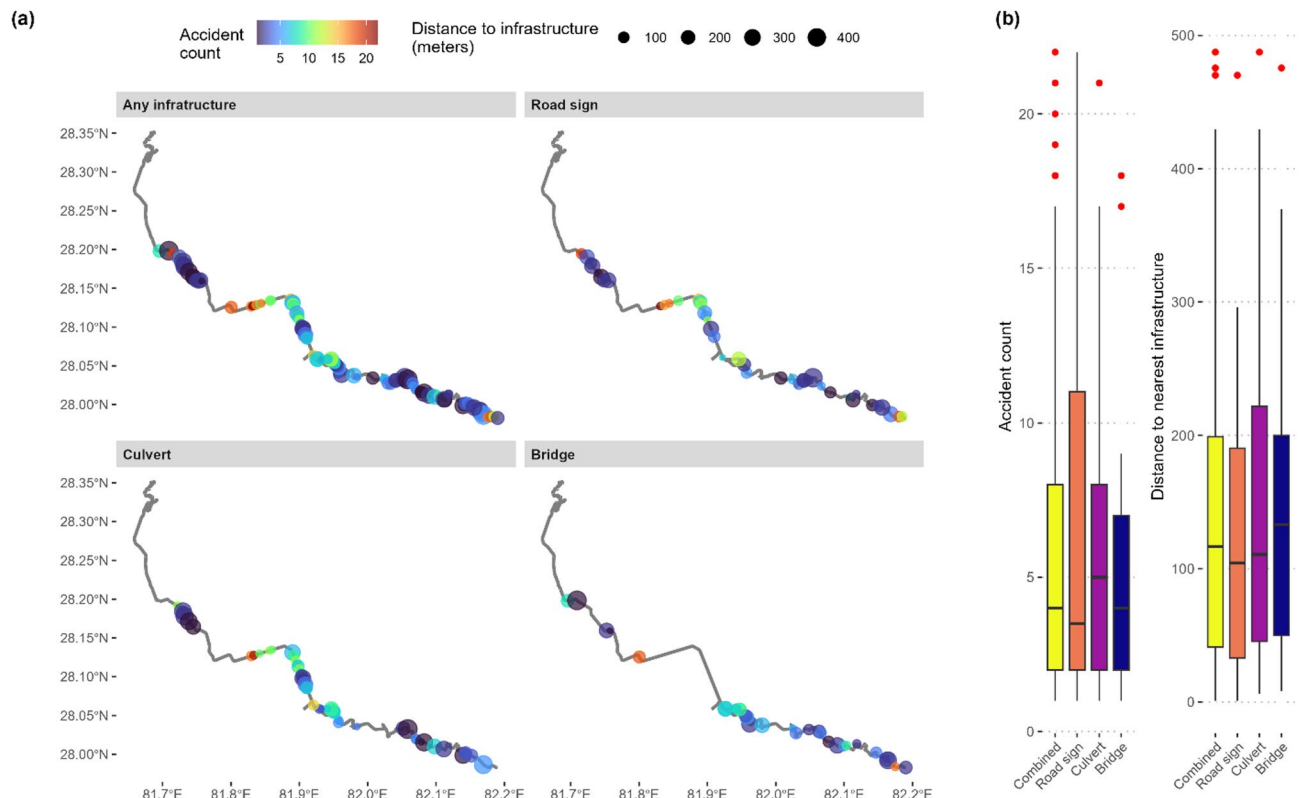


Fig. 6. Impact of highway infrastructures (road signs, bridges and culverts) on wildlife-vehicle collisions (WVC) along the E-W highway in Banke National Park (BaNP).

may be related to changes in animal seasonal behaviours, such as migration, mating, and foraging activities, which can increase movement across roads^{71,72}. Magar, et al.³⁷ also reported similar increased collisions during autumn. This finding might coincide with post-monsoon dispersal and the search for food^{73,74}, whereas the relatively lower number of collisions during summer could be due to reduced animal activity during the hotter months^{75,76}.

Factors influencing WVC

This study identified four key factors significantly associated with the occurrence of WVC: road curvature, crown cover, straightness of the road, and road surface topography. These findings are consistent with other studies, where similar environmental factors have been shown to influence WVC risk^{42,77}. For instance, straight roads have been linked to higher vehicle speeds, which increases the likelihood of collisions with wildlife, as observed in the current study where roads with higher straightness had higher odds of WVC^{78,79}. Similarly, crown cover was found to be a significant predictor of WVC, with areas of less than 25% crown cover experiencing a fourfold increase in collision risk compared to areas with no crown cover. This is likely because reduced crown cover facilitates higher visibility for both wildlife and drivers, allowing for more crossings but also increasing the risk of collisions⁸⁰.

In addition to environmental and anthropogenic factors, temporal surges in human activity—such as festival seasons (e.g., Dashain, Tihar) and peak tourism periods (spring and autumn)—may exacerbate WVC risks. During these times, the traffic volume on the East-West Highway increased substantially because of pilgrimage travel, holiday commuting, and tourist influx. For example, during autumn, which coincides with Nepal's major festivals and post-monsoon tourism, the highest number of WVCs (30% of annual incidents) was recorded. This aligns with studies demonstrating that elevated traffic density during cultural or recreational peaks heightens collision probabilities.

The study also highlighted the role of topography in WVC, with undulating surface areas showing significantly lower odds of WVC than flat areas. This finding aligns with research showing that complex terrain may act as a natural deterrent for wildlife crossings, reducing the frequency of collisions⁷². On the other hand, flat or uniform areas with open visibility may encourage more frequent road crossings, particularly by larger mammals⁸¹.

Impact of infrastructure on WVC

The findings of this study highlight the significant role of road infrastructure, specifically road signs, bridges, and culverts, in influencing the frequency and distribution of WVC in BaNP. The analysis revealed that road signs were associated with the highest median accident counts, with WVC occurring more often closer to these structures (median distance of 50–100 m). This suggests that while intended to alert drivers to potential wildlife

crossings, road signs may accidentally indicate zones where wildlife activity is high, leading to increased collision risk. The presence of road signs in areas with high wildlife movement, such as near water sources or forest edges, may also reflect the failure of drivers to reduce speed or remain vigilant in these zones⁸¹. In contrast, culverts presented the lowest median accident counts and relatively consistent distances from WVC (median ~ 150 m). This finding indicates that culverts may serve as effective mitigation structures by providing safe crossing routes for wildlife. Culverts are known to facilitate the movement of small- to medium-sized animals across roads, reducing the likelihood of collisions^{70,82}. The consistent placement of culverts in areas with high wildlife activity, such as near water sources or forest edges, further enhances their effectiveness in mitigating WVC. These findings align with previous studies that have demonstrated the success of culverts in reducing wildlife mortality on roads⁸³. While showing moderate accident counts, bridges exhibited greater variability in distances from WVC (median ~ 200 m). This suggests that bridges may provide some level of mitigation but are less effective than culverts in reducing collision risk. Bridges are often designed for larger animals or to span significant barriers, such as rivers, but their effectiveness in facilitating wildlife crossings depends on their design and placement⁸⁴. In BaNP, the clustering the WVC near road signs reveals a policy gap: static signs alone are ineffective without measures such as habitat connectivity and driver behaviour changes. In Nepal, policies emphasize signage but lack ecological integration and enforcement. We recommend adaptive approaches, including dynamic signs for animal detection and seasonal speed controls in hotspots.

Conservation implications and mitigation measures

The impact of WVC on the species listed in the IUCN Red List and CITES appendices, including endangered and vulnerable species, calls for urgent conservation intervention. The disproportionate impact on the IUCN Red List species (e.g., endangered golden monitor lizard, *Varanus flavescens*) underscores the urgency of integrating road ecology into global conservation agendas such as the UN Sustainable Development Goals (SDG 15) and the post-2020 Global Biodiversity Framework^{85,86}.

Effective mitigation measures can reduce the occurrence of WVC, as shown by studies in other protected areas. These measures include the construction of wildlife underpasses or overpasses, speed reduction zones in identified hotspots, and enhanced road signage to alert drivers to potential wildlife crossings^{82,83}. In BaNP, targeted interventions at hotspots such as Khairi, Muguwa, and Obhari could significantly reduce WVC incidents. Seasonal measures, such as heightened awareness campaigns during high-risk months (autumn and spring), could also help mitigate the temporal peaks in collisions⁸⁷.

Additionally, promoting road designs that reduce vehicle speeds in wildlife-rich areas, such as incorporating more curves or speed bumps, may further lower collision risks⁸¹. Incorporating wildlife behavioral data into road planning and management, including the sustainable road development of wildlife corridors that funnel animals safely across roads, is also critical^{84,88}. This integrated approach would help safeguard wildlife populations and reduce human-wildlife conflicts, thus contributing to conservation efforts in BaNP and other protected areas^{89,90}. By contextualizing Nepal's experience and applying the results to worldwide efforts to balance development and conservation, this study provides actionable insights for policymakers, ecologists, and engineers working to reconcile human mobility with ecological resilience in an era of unprecedented road expansion^{91,92}.

Conclusion

The study in BaNP adds to the growing body of literature highlighting the adverse effects of roads on wildlife. These findings emphasize the need for immediate conservation actions to mitigate the impact of WVC, especially on species of conservation concern. Identifying WVC hotspots and the environmental factors influencing roadkill risk provides a foundation for developing targeted mitigation strategies. By implementing appropriate interventions, such as the construction of wildlife crossings, traffic calming measures, and public awareness campaigns, the risks posed by roads to wildlife in BaNP can be significantly reduced. These efforts will contribute to the long-term conservation of the Park's biodiversity and the preservation of its vulnerable wildlife populations.

Data availability

All data generated or analyzed during this study are included in this published article [and its supplementary information files].

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Author contributions

NPK and AK: Conceptualization, Methodology, Field work, Investigation, Validation, Data analysis, supervision, writing original draft, Writing review & editing, Funding acquisition, Visualization. KD and AG: Field work, Investigation, Validation, Data analysis, Writing review & editing, Visualization. PG and SP: Investigation, data analysis, Writing review & editing. ML, LN, KOO, XLJ, and RCK: Data analysis, Methodology, writing original draft, Writing review & editing. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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